

Mock Data Science Project:

Title: Advertising Sales Prediction Model

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Date: 15.05.2025

Summary:

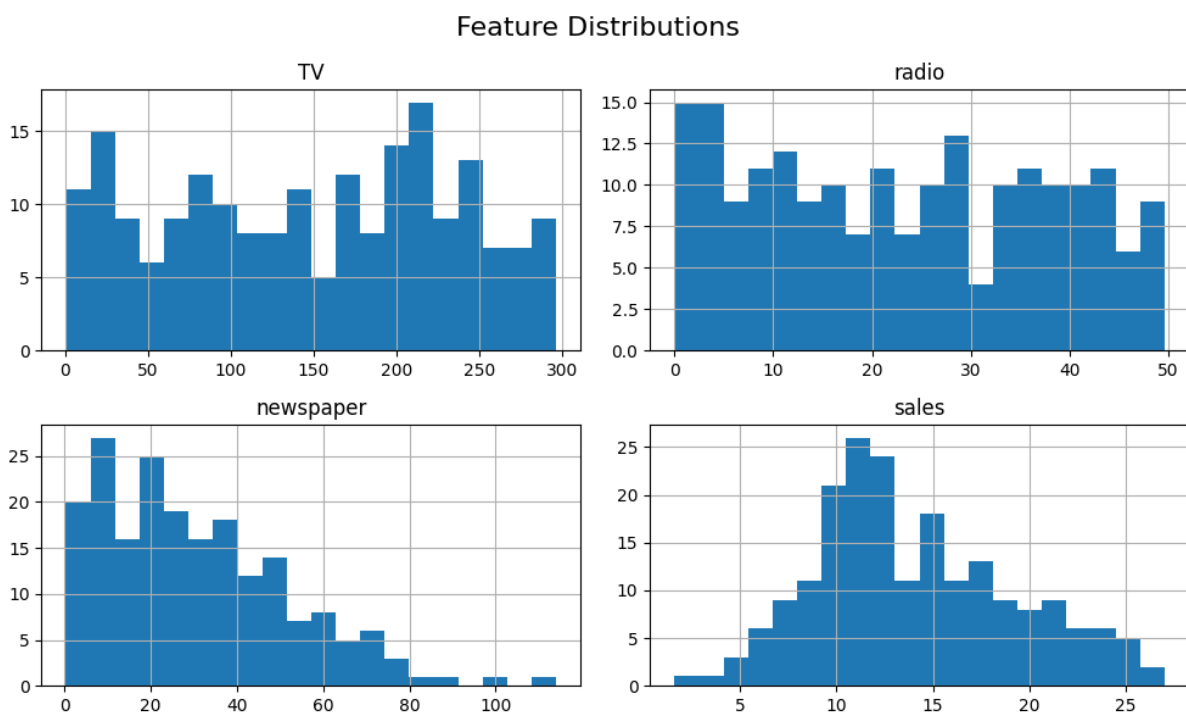
This project aims to build a regression model to predict sales based on advertising data from 3 different aspects: TV, newspaper, and radio. The dataset first cleaned, analyzed, and preprocessed, then a Linear Regression Model was trained on it.

Dataset Description:

- **Source:** [Resources - ISL with R, 1st Edition — An Introduction to Statistical Learning](#)
- **Size:** 200 records, 3 features (TV, radio, newspaper), 1 target (sales).

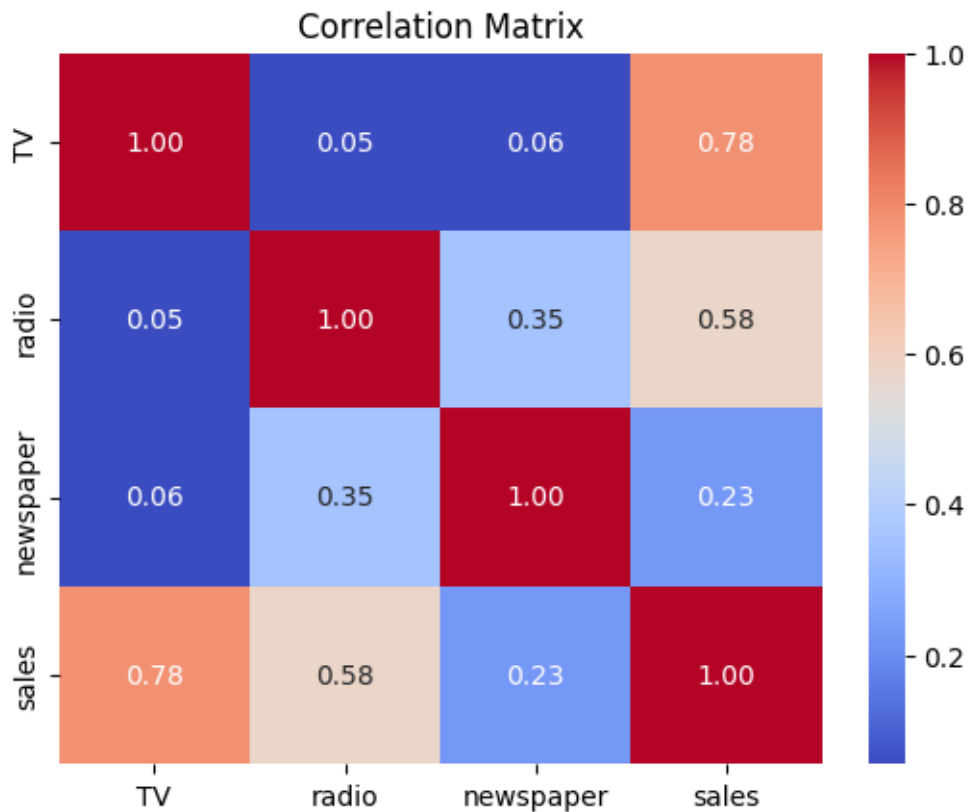
Feature Distributions:

- **TV** and **radio** spending are roughly normally distributed.
- **Newspaper** spending is right-skewed.



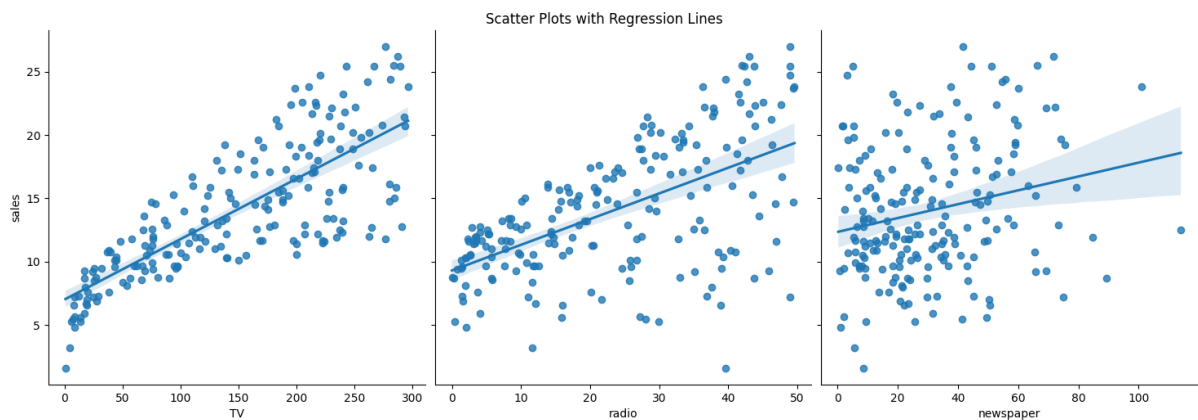
Correlation Analysis:

- **TV** has the strongest correlation with sales (**0.78**).
- **Radio** shows moderate correlation (**0.58**).
- **Newspaper** has the weakest one (**0.23**).



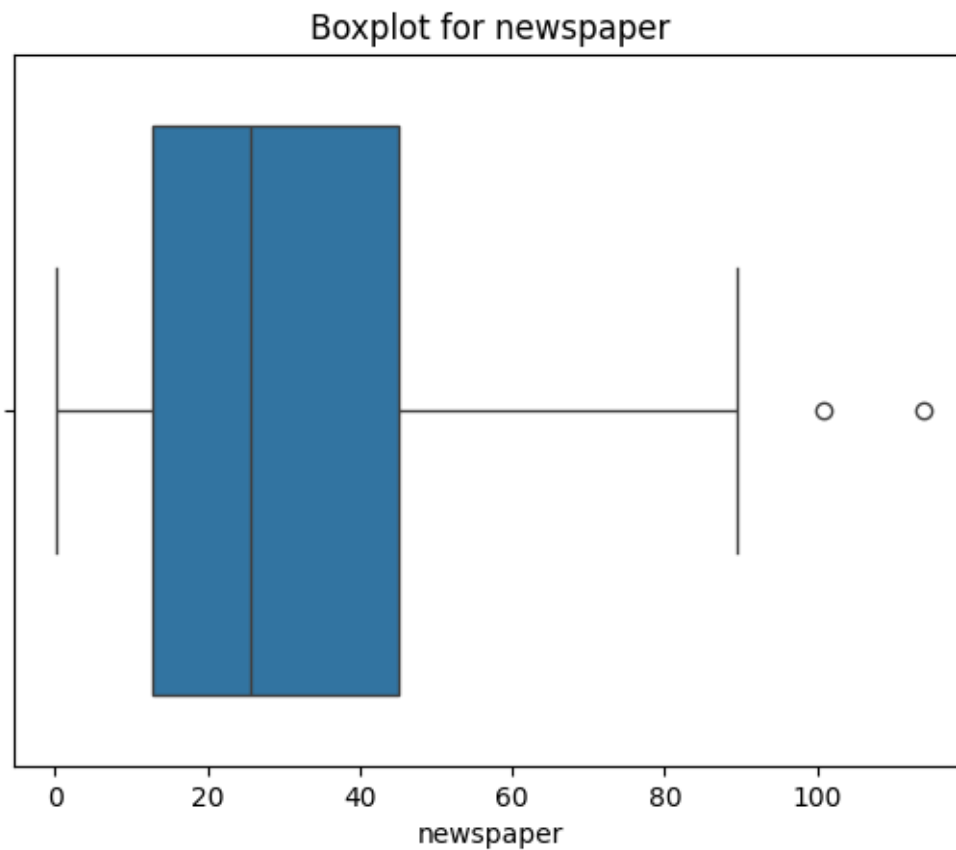
Relationships:

- **TV** and **radio** has good linear trends with sales.
- However, **newspaper** has high variance and weakly predictive.



Outlier Detection:

- Outliers in newspaper were capped to the upper bound (**IQR method**).



Preprocessing:

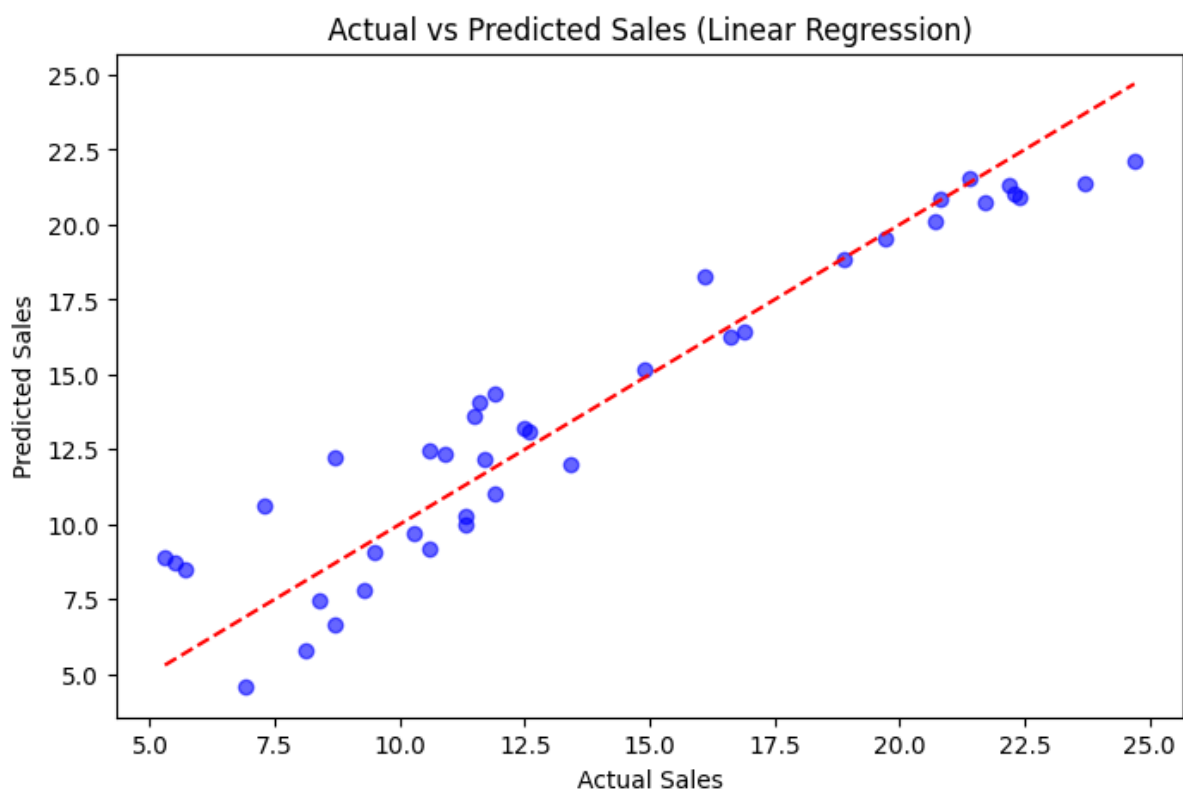
- **Scaling:** Features standardized using StandardScaler.
- **Train-Test Split:** 80% for training, 20% for testing (random_state=42).

Feature Impact:

- TV - 3.830819
- Radio - 2.802285
- Newspaper - 0.057980

Modeling:

- **Algorithm:** Linear Regression
- **Evaluation Metrics:**
 - **RMSE:** 1.781 (lower better).
 - **MSE:** 3.171 (again lower better)
 - **R²:** 0.90 (closer to 1 better).



Key Findings:

- The top predictors for the sales was TV as it has the strongest impact over 'sales' (coefficient ≈ 3.83)
- While newspaper was the least impacting feature for prediction (coefficient ≈ 0.058)